

JANUARY 12 2010

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J. Acoust. Soc. Am. 127, EL30–EL36 (2010)

<https://doi.org/10.1121/1.3280237>



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A simple method to account for size effects in the transfer matrix method

Dilal Rhazi^{a)} and Nouredine Atalla

Department of Mechanical Engineering, GAUS, Université de Sherbrooke, Sherbrooke, Quebec J1K 2R1, Canada
dilal.rhazi@usherbrooke.ca, noureddine.atalla@usherbrooke.ca

Abstract: The transfer matrix method based is extensively used and well-validated for predicting the transmission loss of multilayer structures. However, this method leads to poor results at low frequencies due to the infinite extent assumption it is based on. This paper presents an efficient implementation of a Rayleigh-integral based method to account for the finite size effects. The accuracy of the method is illustrated by various examples.

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PACS numbers: 43.40.Rj, 43.25.Qp, 43.50.Cb [JM]

Date Received: April 8, 2009 **Date Accepted:** December 2, 2009

1. Introduction

The transfer matrix method (TMM) is widely used to predict the airborne transmission loss (TL) of flat infinite structures with added sound packages (Allard, 1993). To account for finite size effects and thus increase the accuracy of the method at low frequencies, Villot *et al.* (2001) proposed a wave approach based on spatial windowing to solve the response of multilayer structures with both airborne and a mechanical excitations. Ghinet and Atalla (2001) proposed a variant based on the Rayleigh integral to account for the size effect. The rationale behind both approaches is to replace the “infinite size” radiation efficiency in the receiving medium, by the radiation efficiency of an equivalent baffled window. The Rayleigh-integral based method, referred to by the finite transfer matrix method (FTMM), has been used by Atalla *et al.* (2006) to solve both the diffuse field transmission loss and absorption of flat multilayer structures. Various experimental examples were presented by the previous authors to corroborate the validity of the proposed methods. However, the main disadvantage of these methods is their high computational cost, especially for a structure-borne excitation. The aim of this note is to present an analytical simplification of the FTMM that allows for a fast and accurate computation of the radiated and transmitted power of multilayered panels.

2. Theory

For the sake of conciseness, the transmission problem is used to illustrate the method. The diffuse field transmission coefficient of a multilayer panel is given by

$$\tau = \frac{\int_0^{\theta_{\text{lim}}} \int_0^{2\pi} \tau_{\infty}(\theta, \varphi) \sin \theta \cos \theta d\varphi d\theta}{\int_0^{\theta_{\text{lim}}} \int_0^{2\pi} \sin \theta \cos \theta d\varphi d\theta}, \quad (1)$$

where $\tau_{\infty}(\theta, \varphi)$ is the transmission coefficient calculated using the TMM in the present context. To account for size effects, the FTMM proposes to replace the infinite size transmission coefficient by a finite size coefficient given by $\tau_f = \tau_{\infty}(\sigma_R \cos \theta)$, where σ_R is homogeneous to radiation efficiency and is given by (Atalla *et al.*, 2006)

^{a)} Author to whom correspondence should be addressed.

$$\sigma_R = \frac{\Re(Z)}{\rho_0 c_0}, \quad (2)$$

with

$$Z = \frac{j\rho_0\omega}{S} \int_S \int_S \exp[-jk_t(x \cos \varphi + y \sin \varphi)] G(M, M_0) \\ \times \exp[jk_t(x' \cos \varphi + y' \sin \varphi)] dx dy dx' dy'. \quad (3)$$

Here S is the surface of the panel, $k_t = k_0 \sin \theta$ with k_0 the acoustic wavenumber, and ρ_0 and c_0 are the density and speed of sound in the receiver medium, respectively. G denotes the half space (baffled) Green's function:

$$G(x, y; x', y') = \frac{\exp[jkR]}{2\pi R}, \quad R = \sqrt{(x-y)^2 + (x'-y')^2}. \quad (4)$$

Note that the impedance Z as given by Eq. (3) only depends on the geometry of the panel, the angle of incidence θ , and heading angle φ . In the following, an analytical reduction in the order of integration in Eq. (3) from 4 to 1 is proposed. Let L_x and L_y denote the length and the width of the structure, respectively. Using the change in variables $\alpha = 2x/L_x$, $\beta = 2y/L_y$, and defining the plate's aspect ratio by $r = L_x/L_y$, Eq. (3) is first written in the form

$$Z = j\rho_0\omega \frac{L_y}{16\pi} \int_0^2 \int_0^2 \int_0^2 \int_0^2 F_n(\alpha, \beta, \alpha', \beta') K(\alpha, \beta, \alpha', \beta') d\alpha d\beta d\alpha' d\beta' \quad (5)$$

with

$$K(\alpha, \beta, \alpha', \beta') = \frac{\exp[-jk_0 R]}{\left[(\alpha - \alpha')^2 + \frac{(\beta - \beta')^2}{r^2} \right]^{1/2}}, \quad (6)$$

$$F_n(\alpha, \beta) = \exp \left[-j \frac{k_r L_x}{2} \left[(\alpha - \alpha') \cos \varphi + \frac{1}{r} (\beta - \beta') \sin \varphi \right] \right], \quad (7)$$

and

$$R = \frac{L_x}{2} \left[(\alpha - \alpha')^2 + \frac{1}{r^2} (\beta - \beta')^2 \right]^{1/2}. \quad (8)$$

To reduce the quadruple integrals in Eq. (5) from 4 to 2, another change in variables is used. Let

$$u = \alpha - \alpha', \quad u' = \beta - \beta', \\ v = \alpha', \quad v' = \beta'. \quad (9)$$

Considering the variable α (the same formula can be written for β), a symbolic form of this change in variable reads

$$\int_0^2 \int_0^2 d\alpha d\alpha' \rightarrow \int_0^2 du \int_0^{2-u} dv + \int_{-2}^0 du \int_{-u}^2 dv. \quad (10)$$

If K and F_n are rewritten in terms of u and u' ,

$$K(u, u') = \frac{\exp\left[-j \frac{k_0 L_x}{2} \left[u^2 + \frac{u'^2}{r^2}\right]^{1/2}\right]}{\left[u^2 + \frac{u'^2}{r^2}\right]^{1/2}}; \quad (11)$$

$$F_n(u, u') = \exp\left[-j \frac{k_t L_x}{2} u \cos \varphi\right] \exp\left[-j \frac{k_t L_x}{2} \frac{u'}{r} \sin \varphi\right].$$

Equation (10) leads to

$$\begin{aligned} & \int_0^2 \left(\int_0^{2-u} dv \right) K(u, u') \exp\left[-j \frac{k_t L_x}{2} u \cos \varphi\right] du \\ & + \int_{-2}^0 \left(\int_{-u}^2 dv \right) K(u, u') \exp\left[-j \frac{k_t L_x}{2} u \cos \varphi\right] du \\ & = 2 \int_0^2 (2-u) K(u, u') \cos\left[\frac{k_t L_x}{2} u \cos \varphi\right] du. \end{aligned} \quad (12)$$

In consequence, the radiation impedance simplifies into

$$Z = j \rho_0 \omega \frac{L_y}{4\pi} \int_0^2 \int_0^2 (2-u)(2-u') K(u, u') \cos\left[\frac{k_t L_x}{2} u \cos \varphi\right] \cos\left[\frac{k_t L_x}{2} \frac{u'}{r} \sin \varphi\right] du du'. \quad (13)$$

Explicitly, Eq. (13) reads

$$\begin{aligned} Z = j \rho_0 \omega \frac{L_y}{4\pi} \int_0^2 \int_0^2 \frac{(2-u)(2-u')}{r_0} & \left[\cos\left(\frac{k_0 L_x r_0}{2}\right) \cos\left(\frac{k_t L_x}{2} u \cos \varphi\right) \cos\left(\frac{k_t L_x}{2} \frac{u'}{r} \sin \varphi\right) \right. \\ & \left. - j \sin\left(\frac{k_0 L_x r_0}{2}\right) \cos\left(\frac{k_t L_x}{2} u \cos \varphi\right) \cos\left(\frac{k_t L_x}{2} \frac{u'}{r} \sin \varphi\right) \right] du du', \end{aligned} \quad (14)$$

with $r_0 = \sqrt{u^2 + (u'^2/r)}$.

To evaluate the radiation efficiency equation (2), only the real part of Z is needed. Making the change in variables, $X=u$ and $Y=u'/r$, the real part of radiation impedance equation (14) becomes

$$\begin{aligned} \Re[Z] = \rho_0 \omega \frac{L_y}{4\pi} \int_0^2 \int_0^{2/r} r & \\ & \times \frac{(2-X)(2-rY) \sin\left(\frac{k_0 L_x r_0}{2}\right) \cos\left(\frac{k_t L_x}{2} X \cos \varphi\right) \cos\left(\frac{k_t L_x}{2} Y \sin \varphi\right)}{r_0} dX dY \end{aligned} \quad (15)$$

A second change of variables is used to work in polar coordinates:

$$\begin{cases} X = R \cos \psi \\ X = R \sin \psi \end{cases} \rightarrow dX dY = R dR d\psi \quad \text{and} \quad R = \sqrt{X^2 + Y^2} = r_0. \quad (16)$$

Using the relation $\cos(\psi - \varphi) = \cos \psi \cos \varphi + \sin \psi \sin \varphi$, Eq. (15) becomes

$$\Re[Z] = \rho_0 \omega \frac{L_y}{4\pi} \int_0^{\pi/2} r F_R(\psi) d\psi, \quad (17)$$

with

$$F_R(\psi) = \frac{1}{4}(F_R(\psi, \alpha_1) + F_R(\psi, \alpha_2) + F_R(\psi, \alpha_3) + F_R(\psi, \alpha_4)) \quad (18)$$

and

$$F_R(\psi, \alpha_i) = \int_0^{R_\theta} (2 - R \cos \psi)(2 - rR \sin \psi) \sin(R\alpha_i) dR, \quad (19)$$

$$\alpha_{1,2} = \frac{L_x}{2}[k_0 \pm k_t \cos(\psi - \varphi)], \quad \alpha_{3,4} = \frac{L_x}{2}[k_0 \pm k_t \cos(\psi + \varphi)].$$

Integral $F_R(\psi, \alpha_i)$ can be performed analytically:

$$F_R(\psi, \alpha_i) = \frac{4}{\alpha_i} + \cos(\alpha_i R_\theta) \left[-\frac{4}{\alpha_i} + 2\frac{R_\theta}{\alpha_i} [r \sin \psi + \cos \psi] + r \cos \psi \sin \psi \left(\frac{2}{\alpha_i^3} - \frac{R_\theta^2}{\alpha_i} \right) \right] \\ + \sin(\alpha_i R_\theta) \left[-\frac{2}{\alpha_i^2} (r \sin \psi + \cos \psi) + \frac{2r}{\alpha_i^2} R_\theta \cos \psi \sin \psi \right] - \frac{2}{\alpha_i^3} r \cos \psi \sin \psi, \quad (20)$$

where

$$\left. \begin{aligned} R_\theta &= \frac{2}{\cos \theta} & \text{if } \theta \leq \theta_l \\ R_\theta &= \frac{2}{r \sin \theta} & \text{if } \theta > \theta_l \end{aligned} \right\} \quad \theta_l = \arctan\left(\frac{1}{r}\right). \quad (21)$$

Finally, the integration limits in Eq. (17) are changed using $\mu = (4\psi/\pi) - 1$ so that a Gauss numerical integration scheme will be used to evaluate the radiation efficiency equation (2):

$$\sigma_R(k_t, \varphi) = \frac{k_0 L_y}{16} \int_{-1}^1 r F(\mu) d\mu. \quad (22)$$

For completeness, note that the imaginary part of radiation impedance can be evaluated using the same approach. It is given by

$$\Im[Z] = \rho_0 \omega \frac{L_y}{4\pi} \int_0^{\pi/2} r F_I(\psi) d\psi, \quad (23)$$

where

$$F_I(\psi) = \frac{1}{4}(F_I(\psi, \alpha_1) + F_I(\psi, \alpha_2) + F_I(\psi, \alpha_3) + F_I(\psi, \alpha_4)), \quad (24)$$

$$F_I(\psi, \alpha_i) = \int_0^{R_\theta} (2 - R \cos \psi)(2 - rR \sin \psi) \cos(R\alpha_i) dR. \quad (25)$$

3. Examples

To illustrate the application of the method, the estimation of the radiation efficiency of a thin plate is first considered. Note that by replacing in Eq. (22) the trace wavenumber k_t by the

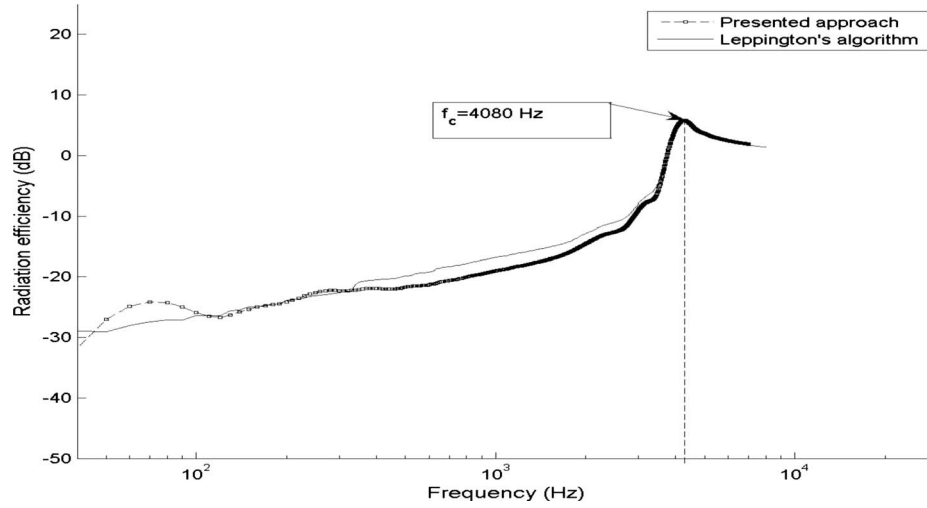


Fig. 1. Radiation efficiency of bare aluminum panel.

bending wavenumber of a structure ($\sqrt{\omega^4 m/D}$ in the case of a thin plate where m is the mass per unit area and D is the bending stiffness) and averaging over heading angles, the radiation efficiency of the structure can be estimated:

$$\bar{\sigma}_R(k_b, \varphi) = \frac{1}{2\pi} \int_0^{2\pi} \sigma_R(k_b, \varphi) d\varphi. \quad (26)$$

Estimation of the radiation efficiency, calculated using Eq. (22), is compared in Fig. 1 to an estimation using Leppington's asymptotic formula (Leppington *et al.*, 1982) for a 3 mm thick 1×1 m² aluminum plate. Good agreement is observed apart from a slight underestimation at mid-frequencies; this is due to the differences theories used for both methods.

Next, to illustrate application of the method to a structure-borne problem, the estimation of the radiated power in the receiver side of a plate-fiber-plate system is considered. The radiated power of an infinite panel can be written in wavenumber space (Cremer *et al.*, 1988):

$$\Pi_{\text{rad}} = \frac{\langle F^2 \rangle}{8\pi^2} \int_0^{2\pi} \int_0^{k_0} |v(\xi, \psi)|^2 \Re Z_\infty(\xi, \psi) \xi d\xi d\psi. \quad (27)$$

Here $\langle F^2 \rangle$ denotes the excitation power spectrum and $v(\xi, \psi)$ denotes the plane wave normal velocity of the radiating panel (calculated using the TMM), and

$$Z_\infty(\xi, \psi) = \frac{k_0 Z_0}{\sqrt{k_0^2 - (\xi_1^2 + \xi_2^2)}} \quad (28)$$

is the radiation impedance of the panel in air (the source domain), $Z_0 = \rho_0 c_0$ being the specific impedance in air. To account for the size effect, it is proposed here to use the same approach as for the transmission loss problem and replace the infinite panel radiation efficiency Z_∞ by Eq. (2). This transforms Eq. (27) into

$$\Pi_{\text{rad}} = \frac{\langle F^2 \rangle}{8\pi^2} \int_0^{2\pi} \int_0^\infty |v(\xi, \psi)|^2 Z_0 \sigma_R(\xi, \psi) \xi d\xi d\psi. \quad (29)$$

Comparing Eq. (27) and Eq. (29), it is important to note that the infinite limit of integration for the wavenumber is important to correctly capture the behavior below the critical frequency of

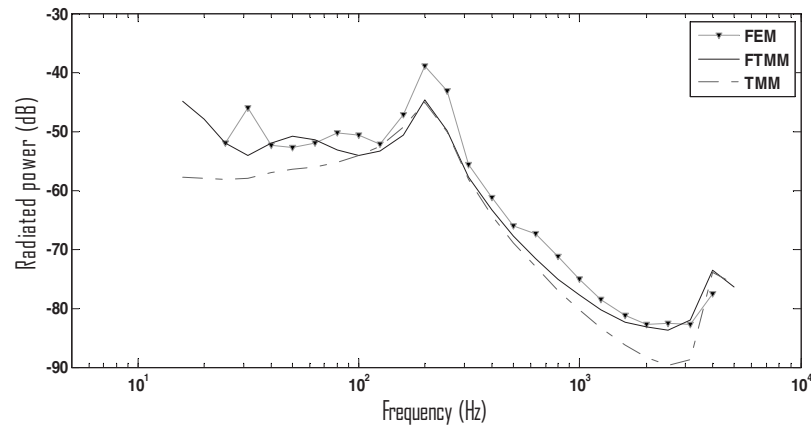


Fig. 2. Radiated power in the receiver side of a double wall structure.

the panel. Because of this double integration, the semi-analytical expression presented in Eq. (17) is important compared to a direct evaluation of Eq. (15) as done previously (Atalla *et al.*, 2006). Figure 2 shows a comparison between the radiated power calculated using Eq. (29) and finite element simulations for a 1×1 m² double wall structure made up a 3 cm thick fiber sandwiched between a 3 mm aluminum plate and 1 mm thick mass layer. The system is excited by a random-position point force (rain on the roof). In both the FTMM [given by Eq. (29)] and the Finite Element Method (FEM), the fiber is modeled as a limp porous layer. To approximate the rain on the roof excitation in the FEM, the results are one-third octave averaged over five randomly chosen locations. Good agreement is observed between the FEM and FTMM. The results obtained using the classical TMM [given by Eq. (27)] are also given to illustrate the importance of size correction for the radiation problem and accuracy of the presented method.

Finally, to illustrate the performance of the method for an airborne excitation, the diffuse field TL of a panel with attached foam is considered. It consists of a flat 1.64 m $\times$ 1.19 m $\times$ 1.016 mm aluminum panel with 7.62 cm thick attached foam. The description of the test and the finite element calculation are given in Atalla *et al.* (2006). The measured transmission loss and predictions with the FTMM are given in Fig. 3. Again, good agreement is observed over the whole frequency range.

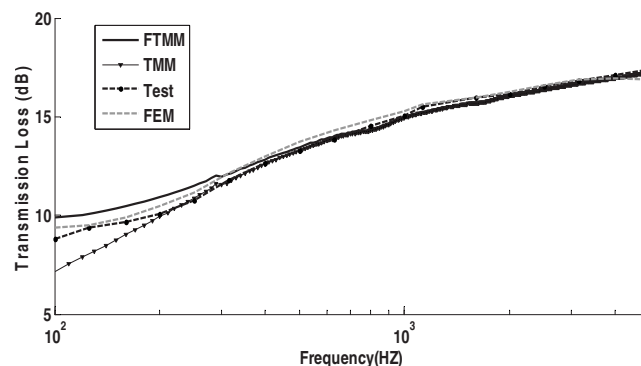


Fig. 3. Transmission loss of a plate-foam system.

4. Conclusion

A simple formulation to evaluate the Rayleigh-integral based size correction for the transfer matrix method is proposed. This simplification allows for an efficient and accurate estimation of the radiated power, radiation efficiency, and transmission loss using the classical TMM. The presented examples illustrate the relevance and accuracy of the presented method in relation to more exact and costly methods, such as the finite element method.

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